Generalized Eigenvalues and the QZ Algorithm

Amp

A generalized eigenvalue is a number λ with $Ax = \lambda Bx$ for two square matrices. It refuses to be an eigenvalue of any single matrix once B is singular, and there is then no $B^{-1}A$ to form, yet the spectrum is perfectly well defined and may contain an infinite eigenvalue. This note is about the object that takes the place of $B^{-1}A$ – the **pencil** $A - \lambda B$, whose eigen-information is stored as a pair (α, β) – and about the algorithm the libraries run to compute it: the QZ (generalized Schur) decomposition, which drives both matrices triangular with one unitary transformation on each side. A three-by-three pencil with a complex pair of eigenvalues and a tunable light mode gives exact eigenvalues to check every step. It shows the finite pair holding machine precision while the second matrix becomes singular, an infinite eigenvalue surviving where $B^{-1}A$ does not exist, and the pair representation holding finite data where the ratio overflows. The note reduces the pencil to Hessenberg-triangular form with real numbers, explains the implicit shift and bulge chase of the iteration, and closes with the library call and how to read its output. Scope: dense, general pencils. Structure preserving methods for Hamiltonian or symplectic pencils, and large sparse problems, are named but not attempted.

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1. An eigenvalue can belong to two matrices at once

The eigenvalue problem every engineer meets first is $Ax = \lambda x$: one matrix, one number, one direction. The object of this note replaces one matrix by two, $Ax = \lambda Bx$, and that single change moves the eigenvalue out of any one matrix and into the **pair**.

Where the pair comes from. A linear descriptor system

$$E\dot{x} = Ax \tag{1}$$

has modes $x(t) = e^{\lambda t}v$ when $Av = \lambda Ev$: the growth rate λ and the shape v are a generalized eigenvalue and eigenvector of the pair (A, E). Eliminating λ leaves a polynomial condition, $\det(A - \lambda E) = 0$, in which the two matrices appear only together. Write the **matrix pencil**

$$A - \lambda B \tag{2}$$

and the condition is that the pencil is singular: the eigenvalue is the λ that makes $A - \lambda B$ lose rank. When $B = I$ this is the ordinary eigenproblem; in general it is not the eigenproblem of $B^{-1}A$ or of any other single matrix, and the rest of the note is about respecting that.

The example, and its exact spectrum. To be able to check every step, take a pencil whose answer we know. Start in a basis where it is easy: an undamped unit oscillator (a rotation block) sitting next to a scalar mode,

$$S_0 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad T_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \delta \end{pmatrix}. \tag{3}$$

The oscillator block contributes $\lambda^2 + 1 = 0$, that is $\lambda = \pm i$, and the scalar pair gives $\lambda = 1/\delta$. So

$$\det(S_0 - \lambda T_0) = (\lambda^2 + 1)(1 - \delta\lambda), \quad \text{spectrum} \quad \{i, -i, 1/\delta\}, \tag{4}$$

and as the mode's "inertia" δ goes to zero its rate $1/\delta$ runs off to infinity. The basis is deliberately a coupling-free one; the pair the algorithm actually receives is the same pencil written in coordinates where the light mode couples to the oscillator. A rotation U of the $(1, 3)$ plane, $U = \begin{pmatrix} c & 0 & -s \\ 0 & 1 & 0 \\ s & 0 & c \end{pmatrix}$ with $c = s = 1/\sqrt{2}$, produces the dense pair

$$A = US_0U^T = \begin{pmatrix} 0.5 & -0.7071 & -0.5 \\ 0.7071 & 0 & 0.7071 \\ -0.5 & -0.7071 & 0.5 \end{pmatrix}, \quad B = UT_0U^T = \begin{pmatrix} 0.75 & 0 & 0.25 \\ 0 & 1 & 0 \\ 0.25 & 0 & 0.75 \end{pmatrix}. \tag{5}$$

(Evaluated at $\delta = 1/2$; δ enters only through T_0.) Because U is orthogonal, $\det(A - \lambda B) = \det(S_0 - \lambda T_0)$: the change of coordinates moves the numbers inside the matrices and leaves the spectrum alone. The script checks Equation 4 at three values of λ to 3.6×10^{-15}, and checks that the computed eigenpairs satisfy $Av = \lambda Bv$ to a relative residual of 2.5×10^{-16}.

δ is the one knob. It tunes the smallest singular value of B, hence how nearly singular the pencil is; it does not move the complex pair. The figure tracks the three eigenvalues as δ shrinks.

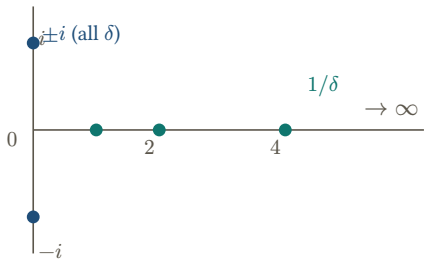


Figure 1: The spectrum of the example as δ falls. The complex pair stays at $\pm i$; the third eigenvalue is $1/\delta$, moving right and reaching infinity at $\delta = 0$. Inspect that only the third eigenvalue moves: tuning δ changes the conditioning of B, not the oscillator.

The point to carry forward: the spectrum has a piece that is finite and fixed, a piece that runs to infinity, and no single matrix in sight that owns either.

2. Inverting B is the wrong first move

If B happens to be invertible, the pencil hides no difficulty at all: multiply $Ax = \lambda Bx$ by B^{-1} and the generalized problem becomes the standard one

$$(B^{-1}A)x = \lambda x. \tag{6}$$

So why not always form $B^{-1}A$ and hand it to the QR algorithm? Two reasons, and the example makes both concrete.

First, B need not be invertible. At $\delta = 0$ the second matrix of Equation 5 is $B = U\,\mathrm{diag}(1, 1, 0)U^T$, of rank 2 and determinant 0: there is no $B^{-1}A$ to form. The pencil, however, is still perfectly regular – $\det(A - \lambda B) = (\lambda^2 + 1)$ is not identically zero – and it has exactly two finite eigenvalues and one infinite one. A library asked for the eigenvalues of the pair returns them:

QUANTITY	$\delta = 1/2$	$\delta = 0$
$\lvert\det B\rvert$	0.375	0
$\mathrm{rank}(B)$	3	2
the ratio route $B^{-1}A$	defined	**not defined**
$\mathrm{eig}(A, B)$	$\{2, i, -i\}$	$\{i, -i, \infty\}$

Table 1: The two routes at the singular endpoint. `B \ A` cannot be formed; the pair's eigenvalues include an infinite one. Printed by `scripts/verify_qz.m`.

Second, even when B is invertible, $B^{-1}A$ is the wrong matrix to perturb. As δ shrinks the entries of $B^{-1}A$ grow like $1/\delta$, and the rounding error of a backward-stable eigensolver is proportional to the norm of the matrix it was given. The **relative** error of the finite pair therefore rises roughly like ε_m/δ, where $\varepsilon_m = 2.2 \times 10^{-16}$ is the unit roundoff, even though the finite pair is a well-conditioned eigenvalue of the pencil. The QZ route, which never forms $B^{-1}A$, holds it at machine precision.

δ	$\kappa(B)$	$\mathrm{eig}(B^{-1}A)$ ERROR	$\mathrm{eig}(A, B)$ ERROR
1	1.0×10^{0}	1.39×10^{-16}	1.11×10^{-16}
$1/2$	2.0×10^{0}	8.95×10^{-16}	1.94×10^{-16}
10^{-2}	1.0×10^{2}	7.25×10^{-15}	2.37×10^{-16}
10^{-4}	1.0×10^{4}	2.67×10^{-13}	2.22×10^{-16}
10^{-8}	1.0×10^{8}	3.98×10^{-9}	4.96×10^{-24}
10^{-12}	1.0×10^{12}	1.08×10^{-6}	1.36×10^{-20}
10^{-16}	2.6×10^{16}	4.14×10^{-1}	2.24×10^{-16}

Table 2: Relative error of the finite pair $\{i, -i\}$, from `scripts/verify_qz.m`. The ratio route $B^{-1}A$ degrades roughly as ε_m/δ; the pencil route stays at machine precision. Inspect the last two columns: the gap is the digits that forming $B^{-1}A$ loses.

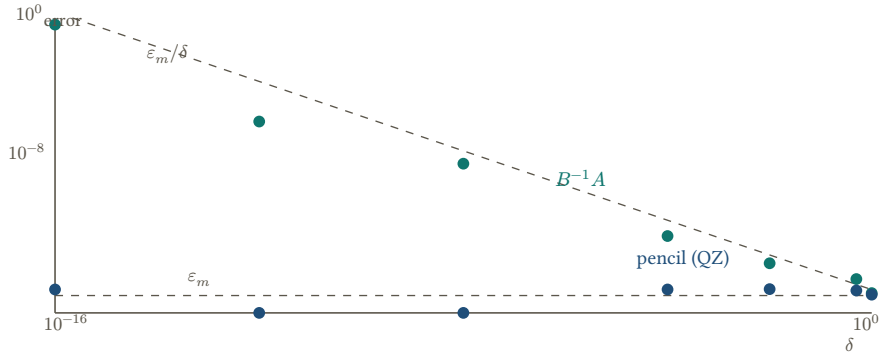


Figure 2: The same table as a picture, on logarithmic axes. The teal series (the ratio route $B^{-1}A$) rises along the steep dashed guide ε_m/δ as δ falls: the digits lost are roughly the digits of $1/\delta$. The navy series (the pencil route) lies on the machine-precision floor, the horizontal dashed line at ε_m. Numbers from `scripts/verify_qz.m`.

What QZ does not buy. It is worth saying plainly, because the table invites the opposite reading: the **large** eigenvalue $1/\delta$ is genuinely ill-conditioned in the pencil, and no algorithm computes it accurately once δ is small. Measured on the same runs, its relative error is 4.4×10^{-13} (ratio route) and 4.7×10^{-13} (pencil route) at $\delta = 10^{-4}$, and 3.3×10^{-5} against 4.7×10^{-5} at $\delta = 10^{-12}$: the two routes agree, because both are now limited by the pencil itself. QZ protects the well-conditioned part of the spectrum; it does not manufacture conditioning that the pencil does not have. This is the same lesson the Riccati note in this repository reaches from the other direction (`Control/qz-riccati/qz-riccati.pdf`, section 7).

There is one more reason the libraries keep the pair rather than the ratio, and it is a representation argument rather than an accuracy argument. LAPACK stores an eigenvalue as α/β and warns that "the quotients α/β may easily over- or underflow, and β may even be zero" [1]. In the decoupled coordinates the third pair is exactly $(1, \delta)$: at $\delta = 10^{-309}$ the two numbers are finite, but their ratio $1/\delta$ overflows a double. The pair survives as data where the ratio does not.

> **Try it:** At $\delta = 0$, what does a `B \ A` solver return, and what does an eigenvalue call on the pair return?

Its answer, visibly below: `B \ A` has no value to compute – a solver either refuses or warns that the matrix is singular to machine precision – while the pair returns the two finite eigenvalues $\pm i$ and reports the third as infinite. Nothing about the **problem** became singular; only the chosen route did.

3. Two triangles, and the pair (α, β)

The classical Schur decomposition makes a single matrix triangular by a unitary similarity; the generalized version makes **both** matrices triangular by a unitary transformation on each side. There exist unitary Q and Z with

$$Q^*AZ = S, \quad Q^*BZ = T, \quad S, T \text{ upper triangular} \tag{7}$$

in complex arithmetic; for real A, B there is a real version in which T is upper triangular and S is only **quasi**-upper-triangular, with 2×2 blocks where the spectrum is complex. This is the generalized Schur, or QZ, decomposition. (Some texts write the factors the other way round, $Q^*AZ = T$ and $Q^*BZ = S$; only the naming differs. This note calls the first matrix's image S and the second's T, so the pairs run (S_{ii}, T_{ii}).)

Why triangles are enough is a one-line determinant. Since Q and Z are unitary, $\det(Q)\det(Z^*) = c$ has unit modulus, and $S - \lambda T$ is triangular, so

$$\det(A - \lambda B) = \det(Q)\det(S - \lambda T)\det(Z^*) = c\prod_i (S_{ii} - \lambda T_{ii}). \tag{8}$$

A nonzero constant c does not move the roots, so the degree-n polynomial on the left has exactly the roots of the product: n linear factors read off the diagonals of S and T. Writing each factor as a pair,

$$(\alpha_i, \beta_i) = (S_{ii}, T_{ii}), \quad \lambda_i = \alpha_i/\beta_i \quad (\beta_i \neq 0), \quad \text{and} \quad \lambda_i = \infty \quad (\beta_i = 0, \alpha_i \neq 0), \tag{9}$$

we have the whole spectrum, finite and infinite, without a single division that could fail. This is the representation the rest of the note uses: an eigenvalue is not a number the algorithm converges to, it is a **pair** on the diagonal, and only the user's final step (if any) turns it into a ratio. The pair is also exactly what LAPACK returns, as α, β. The factorization also groups the spectrum: because S and T are triangular (quasi-triangular in the real form), the leading k columns of Z together with the leading k of Q already form a closed block carrying the first k pairs and nothing else, provided k does not cut across a 2×2 block. Section 6 defines what those columns span – a **deflating subspace** on each side, a pair of spaces rather than one – and ordering is what chooses which pairs sit in that leading block.

Read the example. For $\delta = 1/2$ the complex QZ of Equation 5 is essentially diagonal; the script's run gives the diagonal pairs

$$(1, 0.5) \to 2, \quad (i, 1) \to i, \quad (-i, 1) \to -i, \tag{10}$$

and the transformation is unitary and exact to machine precision: $\|Q^*Q - I\| = 5.3 \times 10^{-16}$, $\|Q^*AZ - S\| = 8.1 \times 10^{-16}$. At $\delta = 0$, where B has lost its rank, the same call returns

$$(i, 1) \to i, \quad (-i, 1) \to -i, \quad (-1, 0) \to \infty. \tag{11}$$

The third pair is the infinite eigenvalue: $\beta = 0$ is not a failure, it is the answer.

The real form, and a trap. Real drivers keep the arithmetic real and therefore cannot put a complex eigenvalue on the diagonal. They return a 2×2 block instead. For $\delta = 1/2$ the real QZ gives

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{12}$$

whose 2×2 block $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ pairs with the identity block of T to give $\lambda^2 + 1$, i.e. the pair $\pm i$. The trap is the obvious mistake: **do not read the diagonal of a 2×2 block as two eigenvalues.** Its diagonal is $(0, 0)$ here, and $(0, 0)$ would suggest a double eigenvalue at zero. A complex pair is the generalized eigenvalue of the whole block, and a library reports it as a block, or as the α_i, β_i list with the imaginary part flagged – never as two independent diagonal ratios.

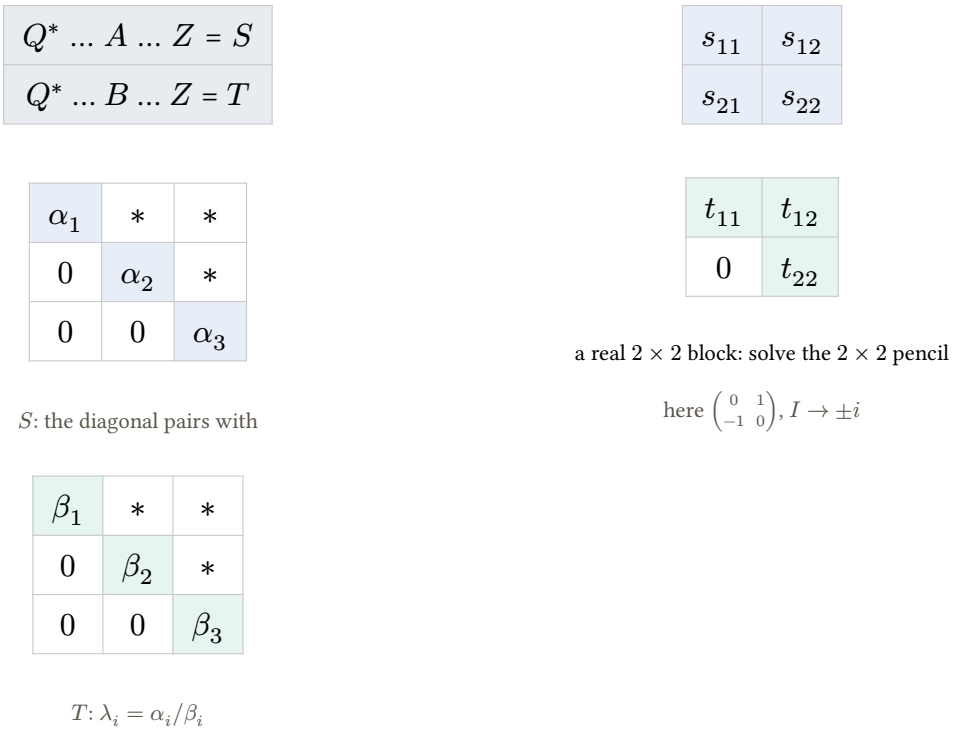


Figure 3: The generalized Schur form. Left: the shape of the factorization Equation 7 – a unitary transformation from each side, both matrices triangular, and the spectrum on the diagonals. Right: the alternative reading the algorithm can return, a 2×2 block whose generalized eigenvalues are a complex pair. Inspect that only the diagonal (left) or the block (right) carries eigenvalues; everything above is coupling and carries none.

4. Reducing the pencil without losing the triangle

The QZ decomposition Equation 7 is found in two movements. The first is a fixed, finite amount of work – a reduction to a special form – and the second is an iteration on that form. This section is the reduction; the next is the iteration. The reduction is where a pencil, unlike a single matrix, needs a two-sided touch, and it is the one step worth seeing in full.

Why reduce at all. Iterating directly on a full pencil would cost $O(n^3)$ per step. If instead A is upper Hessenberg (zero below the first subdiagonal) and B is upper triangular, each iteration step touches only $O(n^2)$ entries, and the whole cost is dominated by the one-time $O(n^3)$ reduction. It is the same economics as reducing a matrix to Hessenberg form before the QR algorithm. The reduction's other job is to expose the zero structure that tells the iteration when a piece has converged.

Phase one: triangularize B. A QR factorization of B from the left, $B = Q_0 R$, makes $R = Q_0^T B$ triangular; applying the same Q_0^T to A leaves the pencil equivalent, because

$$Q_0^T (A - \lambda B) = (Q_0^T A) - \lambda (Q_0^T B). \tag{13}$$

For the example at $\delta = 1/2$ this is a two-Householder step and gives B triangular and A still full:

$$A_1 = \begin{pmatrix} -0.3162 & 0.8944 & 0.3162 \\ -0.7071 & 0 & -0.7071 \\ -0.6325 & -0.4472 & 0.6325 \end{pmatrix}, \quad B_1 = \begin{pmatrix} -0.7906 & 0 & -0.4743 \\ 0 & -1 & 0 \\ 0 & 0 & 0.6325 \end{pmatrix}. \tag{14}$$

Phase two: Hessenbergize A without spoiling B. Now A_1 still has an entry below its band – $A_1(3,1) = -0.6325$ – and it must go. The way to remove it is a rotation of rows 2 and 3 from the left. But that same rotation acts on B_1, whose rows 2 and 3 then mix; the zero at $B_1(3,2)$ fills in:

$$B_1 = \begin{pmatrix} -0.7906 & 0 & -0.4743 \\ 0 & -1 & 0 \\ 0 & 0 & 0.6325 \end{pmatrix} \quad \to \quad \begin{pmatrix} -0.7906 & 0 & -0.4743 \\ 0 & 0.7454 & -0.4216 \\ 0 & -0.6667 & -0.4714 \end{pmatrix}. \tag{15}$$

The fill is exactly one entry, at $(3,2)$, just below the diagonal. It is repaired by a rotation of **columns** 2 and 3 from the right, which zeroes that entry. The repair does not undo the left rotation: the left rotation cleared column 1 of A, the right rotation touches columns 2 and 3, and the two index sets are disjoint. That is the whole trick, and it is why the reduction is **paired**:

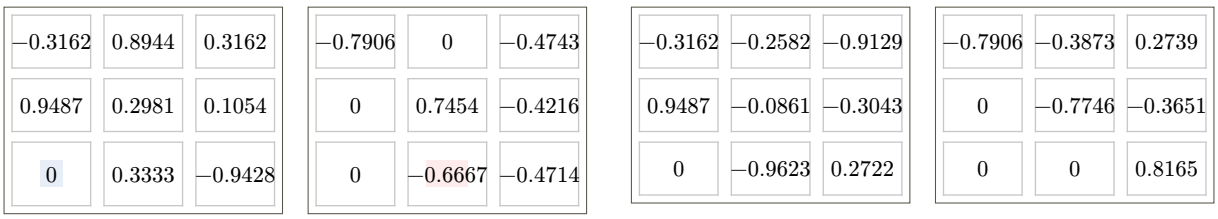


Figure 4: The paired rotations of phase two, on the running example from `scripts/qz_sweeps.m`. Panel 1: after the left rotation, A's entry $(3,1)$ is gone (navy) and B has gained the single entry $(3,2)$ (rose). Panel 2: the right rotation clears $B(3,2)$ and leaves A's zero alone, because it operates on columns 2 and 3 while the zero is in column 1. The result is Hessenberg over triangular.

The general pattern, and why rotations rather than reflectors. Phase two walks down the columns of A and, within each column, from the bottom up, using a **Givens rotation of two adjacent rows at a time**. A Householder reflector on rows $k+1:n$ would zero the whole sub-column $A(k+2:n,k)$ in one blow – and would also turn those rows of B into a dense block below its diagonal, so the repair would stop being local and would spread. A rotation of rows $(i-1,i)$ that clears $A(i,k)$ does the opposite: it fills exactly one entry of B, the $(i,i-1)$ entry just below its diagonal, and the paired rotation of columns $(i-1,i)$ clears it. The cleared entry of A sits in column $k \le i-2$, so the column rotation, acting on columns $i-1$ and i, never refills it. Every left rotation has exactly one right partner, and the whole reduction is a product of such rotations – the form the libraries implement [2].

The script verifies the example end to end: after the two phases, A is Hessenberg and B triangular to 4.1×10^{-17}, the accumulated Q, Z are unitary to 9.2×10^{-16}, and $Q^T AZ$, $Q^T BZ$ reproduce the reduced pair to 6.0×10^{-16} and 7.8×10^{-16}. The reduction has cost $O(n^3)$ and has changed no eigenvalue.

5. The iteration: a shift, a bulge, and a chase

The iteration is the part that actually drives the pencil to triangular form. It is the QR algorithm's step, transplanted to the pencil and done without ever forming a quotient.

The QR step it generalizes. For a single Hessenberg matrix H, one QR step chooses a scalar shift μ – usually an eigenvalue of the trailing 2×2 block – forms the first column of $H - \mu I$, applies a Householder transformation that sends it to a multiple of e_1, and then chases the **bulge** the Householder left behind down the band with further rotations until the Hessenberg shape is restored. Convergence shows up as a vanishing subdiagonal entry, which deflates the problem into two smaller ones. The QZ step is this step applied to the pencil, in the sense that it mirrors one QR step on AB^{-1} – but AB^{-1} is never formed, precisely so that the argument of section 2 does not arise.

The step, and where the bulge comes from. Take the pencil in Hessenberg-triangular form with T invertible. A real driver's step is the one a double-shift QR step would take on the single matrix $M = HT^{-1}$ – which is upper Hessenberg, and which the algorithm never forms. The two shifts σ_1, σ_2 are the eigenvalues of the trailing 2×2 **pencil**, which are also the eigenvalues of M's trailing block. The step begins with the first column of $(M - \sigma_1 I)(M - \sigma_2 I)$; since $M - \sigma I = (H - \sigma T)T^{-1}$, that column is $(H - \sigma_1 T)T^{-1}(H - \sigma_2 T)T^{-1}e_1$. It has only three nonzero entries, because M is Hessenberg and $(M - \sigma_2 I)e_1$ is supported on two coordinates, and it can be assembled from a handful of leading entries of H and T without ever forming T^{-1}; LAPACK builds exactly this 3-vector [3]. A single 3×3 Householder sends it to a multiple of e_1. Applying that Householder from the left is exactly the situation at the end of section 4 – it puts entries below the band – except that now the offending entries are produced by the **shift**, in a patch three wide. The same paired right rotations repair T, and the leftover in H is a **bulge** that is chased one step down the band by the same left/right couple, until it falls off the bottom.

Why two sides are legitimate. The single-matrix QR step is a similarity, $Q^T H Q$; the QZ step is an **equivalence**, $Q^T(A - \lambda B)Z$, with generally different Q and Z. That does not break the implicit Q theorem; it is the theorem applied to the pencil. With $M = HT^{-1}$ (T nonsingular), $Q^T H Z = S$ and $Q^T T Z = P$ together give

$$Q^T M Q = S P^{-1}, \tag{16}$$

an ordinary **similarity** of the single matrix M into upper Hessenberg form. So Q is the orthogonal factor of a Hessenberg reduction of M and is fixed, up to column signs, by its first column Qe_1 – the usual implicit Q theorem – while Z is then forced by the requirement that $Q^T T Z$ be triangular, that is, Z is the orthogonal factor of an RQ factorization of $Q^T T$. The right transformations are the price of never forming T^{-1}; they add no free parameter. Two consequences of the same identity: the shift has to reach the pencil through T^{-1} (so it is the first column above, not $(H - \sigma_1 T)(H - \sigma_2 T)e_1$, which points elsewhere), and the argument assumes T nonsingular – a zero diagonal of T, an infinite eigenvalue, is deflated before it applies.

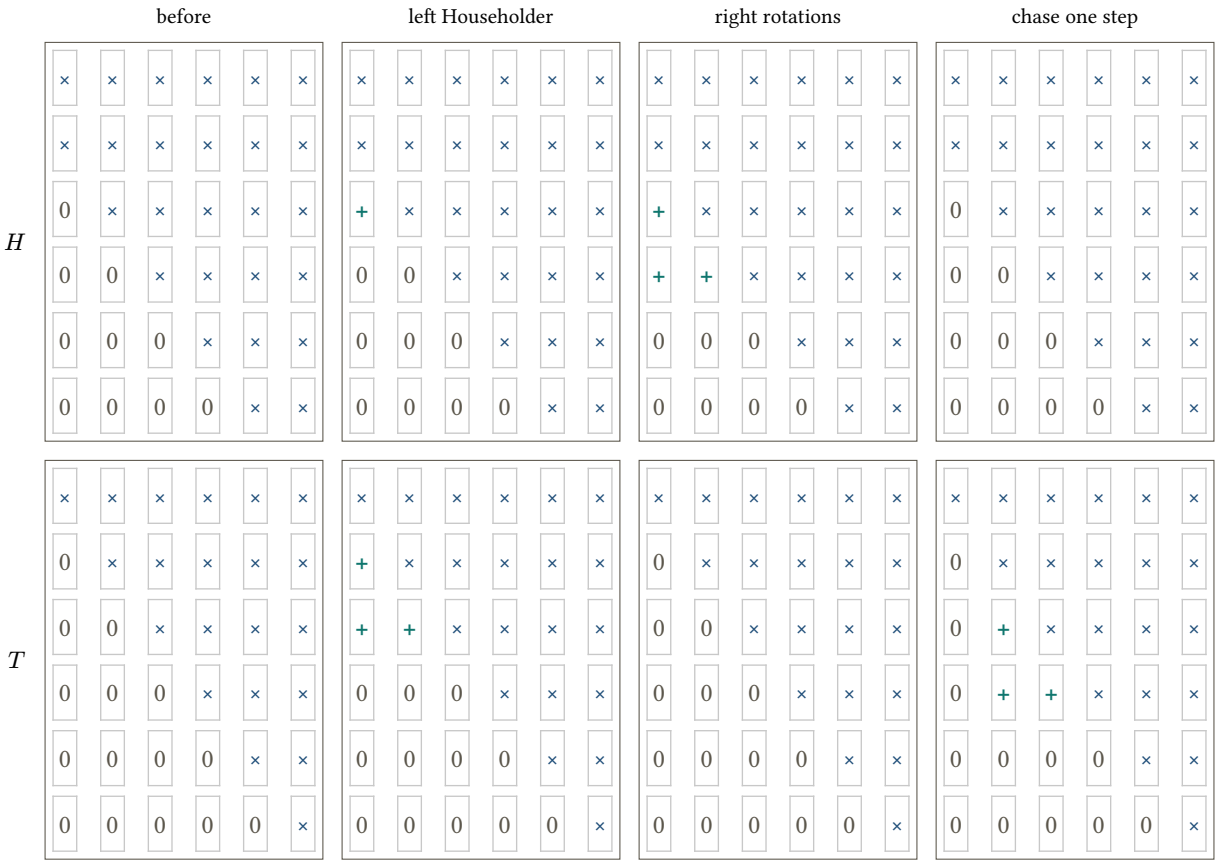


Figure 5: The chase, schematically, on an $n \times 6$ pencil (the pattern a real double-shift step produces). $\times$ is a nonzero, 0 a structural zero, and $+$ an entry that violates Hessenberg-triangular form – the bulge. The left Householder spoils both matrices; the paired right rotations restore T and push the bulge into H; the left rotations restore H and push what is left into T; repeating walks the bulge down and off the bottom. This is the paired rotation of section 4 again, now applied to a bulge rather than to a fixed entry.

STAGE OF ONE SWEEP	BULGE MAGNITUDE
shift from the trailing 2×2 pencil	$\mu = 0.5 + 0.5i$
left Householder: T gains $(2, 1)$	0.7276
right rotation: H gains $(3, 1)$	0.8333
left rotation chases it: T gains $(3, 2)$	0.6623
paired right rotation: bulge gone	0

Table 3: One sweep on the reduced $\delta = 1/2$ pair, from `scripts/qz_sweeps.m`. The shift $\mu = 0.5 + 0.5i$ is the trailing pencil's eigenvalue nearest H_{33}/T_{33}; the numbers are the magnitudes of the moving entry at each stage. This is the complex single-shift form of the step; a real driver takes both trailing eigenvalues at once, which is why its bulge is three wide and it can stay in real arithmetic.

Convergence and deflation. After each sweep the algorithm inspects the subdiagonal coupling: when $|H(i+1, i)|$ is negligible beside its neighbouring diagonal entries of H **and** T, that entry is set to zero and the pencil splits into an upper and a lower block, each solved on its own. This is deflation, the same idea as in the QR algorithm. A zero on the diagonal

of T is handled separately: it means an infinite eigenvalue, and if it is not at the bottom it is chased out of the way (a diagonal version of the same rotation pattern) so that the iteration can continue on the finite part. Running the script's single-shift sweep to convergence on the reduced $\delta = 1/2$ pair takes **seven** sweeps and returns the spectrum to full precision: the computed eigenvalues are $\{2, -i, +i\}$, and the pair is triangular to 2.1×10^{-25} below A's band and 7.4×10^{-18} below the diagonal of B.

The implementation in `scripts/qz_sweeps.m` covers the reduction of section 4 in full and one sweep of the form above; the schematic Figure 5 shows the real double-shift variant that library drivers use, whose inner rotations are the same pair. The note is explicit about which is which: the numbers are from executed code, the wide-bulge pattern is the documented library variant [4], and the implicit double shift itself is left to the library.

6. Using it

The recipe, and then where to find it. Everything below is a dense $O(n^3)$ computation.

1. **Assemble the pencil.** Whatever produces the problem – a descriptor system, a quadratic eigenvalue problem, a Riccati equation – hand the algorithm two matrices A and B, not their product and not an inverse.
2. **Balance** (the driver does this for you). Balancing improves the accuracy of badly scaled pencils by permutations and two-sided diagonal scaling before the iteration [5].
3. **Reduce** to Hessenberg-triangular form, then **iterate** with the implicit shift and chase, deflating as subdiagonals vanish, exactly as sections 4 and 5 describe.
4. **Read the pairs** (α_i, β_i) from the diagonals. The finite eigenvalues are α_i/β_i; $\beta_i = 0$ is an infinite eigenvalue. In the real form, a complex conjugate pair is a 2×2 block, not two diagonals.
5. **Order**, if you want a particular subset first. `ordqz` / `dgges` take a selection rule and move the chosen pairs to the leading block by further unitary transformations – the step the Riccati note uses to put the stable eigenvalues first. The move is a sequence of **adjacent-block swaps**: swapping two real 1×1 blocks is one Givens rotation, while swapping a block that holds a complex pair requires solving the small generalized Sylvester equation $S_{11}R - LS_{22} = \gamma S_{12}, T_{11}R - LT_{22} = \gamma T_{12}$ and building two orthogonal factors from its solution, all applied to both sides so the Schur form is restored with the blocks exchanged [6]. Bubbling the chosen pairs to the front one swap at a time leaves the leading columns of the accumulated Q and Z spanning the corresponding left and right deflating subspaces; a swap that would be too ill-conditioned is reported rather than performed silently.

What a deflating subspace is. The ordering step named an object the drivers compute but the note has not yet defined. For a pencil, the analogue of an invariant subspace is a **pair** of subspaces of equal dimension, one on each side [7]:

$$\dim X = \dim Y = k, \quad AX \subseteq Y, \quad BX \subseteq Y, \tag{17}$$

where AX means $\{Ax : x \in X\}$. Equivalently, with basis matrices X_1 of X and Y_1 of Y, there are $k \times k$ matrices with $AX_1 = Y_1 A_{11}$ and $BX_1 = Y_1 B_{11}$, so the pencil restricted to the pair is the small regular pencil (A_{11}, B_{11}) and its eigenvalues are a subset of the spectrum of (A, B) [8]. The pair is called **deflating**, and its two members are the **right** subspace X and the **left** one Y – the names subsection 6.1 uses for R and L.

Two readings make the definition concrete. Put $B = I$: then $BX \subseteq Y$ is free, Y can be taken to be X, and the condition collapses to $AX \subseteq X$, an ordinary **invariant subspace** – one matrix is the special case. And a single direction is never enough on its own: a generalized eigenvector w, one with $Aw = \lambda Bw$, gives the pair $(X, Y) = (\mathrm{span}(w), \mathrm{span}(Aw))$, whereas an eigenvector of A alone satisfies $AX \subseteq X$ but in general sends BX somewhere else entirely. What the pair buys is the deflation itself: given X and Y, orthonormal bases of them can be completed to unitary Q and Z whose leading k columns hold (A_{11}, B_{11}) and whose trailing columns hold the rest, because then Q^*AZ and Q^*BZ are block upper triangular. That is the certificate the algorithm tests: $S_{k+1,k} = 0$ means the leading columns already have the

property, the pencil splits, and each half is solved on its own, which is the deflation of section 5 [9], [10].

Measured. On the example's pair $\{i, -i\}$ at $\delta = 1/2$ the script reproduces all of it: $\dim(AX + BX) = 2 = \dim X$, the induced 2×2 pencil with eigenvalues $\{i, -i\}$, then orthonormalization with $\|Q^*Q - I\| = 5.0 \times 10^{-16}$ and lower-left blocks 3.1×10^{-16} and 2.6×10^{-16} in Q^*AZ and Q^*BZ, the leading block holding $\{i, -i\}$ and the trailing one holding 2. The same script tests the definition on a tiny pencil that has nothing hidden in it, $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, with eigenvalues $1 \pm i$: an eigenvector of A gives $\dim(AX + BX) = 2 \neq 1$ and is therefore **not** a deflating direction, while a generalized eigenvector gives 1, and $\operatorname{span}(Aw)$ sits $24.1°$ away from $\operatorname{span}(w)$ – the two members of the pair are genuinely different subspaces.

One trap hides in the running example: there B acts as the identity on the $\{i, -i\}$ block ($\|BX_1 - X_1\| = 2.2 \times 10^{-16}$), so that particular pair has $X = Y$ and a one-matrix intuition would appear to survive. The tiny pencil is the honest test of the two-sided definition.

6.1. Moving a block to the front: the swap in detail

The drivers move a selected pair by **adjacent-block swaps** and bubble it forward. Write the two adjacent diagonal blocks as (S_{11}, T_{11}) of size n_1 and (S_{22}, T_{22}) of size n_2, each 1 or 2, with coupling S_{12}, T_{12}. A swap is a pair of orthogonal matrices Q_1, Z_1 with

$$Q_1^T \begin{pmatrix} S_{11} & S_{12} \\ 0 & S_{22} \end{pmatrix} Z_1 = \begin{pmatrix} \tilde{S}_{11} & \tilde{S}_{12} \\ 0 & \tilde{S}_{22} \end{pmatrix}, \quad Q_1^T \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix} Z_1 = \begin{pmatrix} \tilde{T}_{11} & \tilde{T}_{12} \\ 0 & \tilde{T}_{22} \end{pmatrix}, \tag{18}$$

where the leading block carries the eigenvalues the trailing block had. Only this small block is touched directly; the same Q_1, Z_1 are then applied to the coupling rows and columns of the rest of the pencil.

Two real 1×1 blocks. Form the two scalars $F = s_{22}t_{11} - t_{22}s_{11}$ and $G = s_{22}t_{12} - t_{22}s_{12}$, take the Givens rotation (c, s) that sends (F, G) to a multiple of e_1, and apply it from the right to columns 1 and 2. A second rotation – built from (s_{11}, s_{21}) if $|s_{22}| \geq |t_{22}|$ and from (t_{11}, t_{21}) otherwise – is applied from the left to restore triangularity. When $T = I$ this is exactly the standard Schur reordering of a single matrix.

The Sylvester equation, briefly. An equation $AX - XB = C$ with known A, B, C and unknown X is a **Sylvester equation**: linear in X, with the scalar prototype $ax - xb = c$, whose solution is $x = c/(a - b)$. Vectorizing turns it into a linear system of size n^2, and the operator $X \mapsto AX - XB$ has eigenvalues $\lambda_{i(A)} - \lambda_{j(B)}$; so it has one unique solution exactly when A and B share no eigenvalue, and the smallest such difference measures how ill-conditioned that solution is. In the swap, X is what turns coupling into a change of basis: for the 2×2 case above, $x = c/(a - b)$ is exactly the quantity the rotation angle is built from. The pencil doubles the bookkeeping: it is transformed on both sides, so there is a right deflating subspace (parametrized by R) and a left one (parametrized by L), and $Sv = \lambda Tv$ couples the two matrices into the two equations below. The blocks can be exchanged only when their spectra are disjoint, and the **size** of the solution is precisely the swap's conditioning – the quantity the threshold test below checks.

Why the equation has that form. The swap wants a basis whose first directions are the invariant directions of the block that is moving up, and writing such a direction as a **graph** over the block structure is what produces the equation. In the scalar case $M = \left(\begin{smallmatrix} a & c \\ 0 & b \end{smallmatrix}\right)$, let the direction belonging to the eigenvalue b be $v = \left(\begin{smallmatrix} -x \\ 1 \end{smallmatrix}\right)$, with x unknown. Imposing $Mv = bv$ gives $-ax + c = -bx$, that is

$$ax - xb = c. \tag{19}$$

The Sylvester equation **is** the eigenvector equation. In the block case $M = \left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$ the graph is $\left(\begin{smallmatrix} -X \\ I \end{smallmatrix}\right)$, whose bottom block forces the invariant action to be B:

$$\begin{pmatrix} A & C \\ 0 & B \end{pmatrix}\begin{pmatrix} -X \\ I \end{pmatrix} = \begin{pmatrix} -X \\ I \end{pmatrix}B, \quad \text{that is} \quad -AX + C = -XB, \tag{20}$$

which is $AX - XB = C$. The two terms are matched because the graph is acted on from the left by A and from the right by its own block B; a **similarity** – the same basis on both sides – would instead give the commutator $AX - XA$, the Lyapunov equation of control theory. For a pencil the direction satisfies $Sv = \lambda Tv$ for **two** matrices, so the single invariance condition becomes one per matrix, exactly the two equations above; and the two-sided transformation is why there is a right unknown R and a left one L. The script checks both readings on tiny examples: $Mv = bv$ exactly for $x = -0.3333$, and $M\left(\begin{smallmatrix} -X \\ I \end{smallmatrix}\right) = \left(\begin{smallmatrix} -X \\ I \end{smallmatrix}\right)B$ to 2.2×10^{-16}.

From the solution to the rotation. The solution parametrizes the invariant direction, and the swap is that direction made into a basis. In the scalar case the direction is $v = \left(\begin{smallmatrix} -x \\ 1 \end{smallmatrix}\right)$, and the orthogonal matrix whose first column is $v/\|v\|$ is a rotation,

$$G = 1/\sqrt{1 + x^2}\begin{pmatrix} -x & -1 \\ 1 & -x \end{pmatrix}. \tag{21}$$

Because $Mv = bv$, multiplying the first unit vector gives $G^T MGe_1 = be_1$; hence the first column of $G^T MG$ is $(b, 0)$ and the diagonal comes out swapped:

$$G^T MG = \begin{pmatrix} b & * \\ 0 & a \end{pmatrix}. \tag{22}$$

For $a = 1, c = 1, b = 4$ this gives $x = -0.3333$, $\left\|G^T G - I\right\| = 1.1 \times 10^{-16}$, and a diagonal of $(4, 1)$. The block case for a single matrix is the same move: the invariant subspace is the graph $\left(\begin{smallmatrix} -X \\ I \end{smallmatrix}\right)$, orthonormalizing it supplies the leading columns of G, and $G^T MG$ swaps the blocks – the script finds leading eigenvalues $\{4, 5\}$ (the old B), trailing $\{1, 3\}$ (the old A), and a lower-left block of 6.2×10^{-16}.

A pencil cannot do this with one rotation. Its generalized eigenvalues are invariant under an **equivalence** $Q_1^T(S, T)Z_1$ with $Q_1 \neq Z_1$, and a single similarity would in general neither keep T triangular nor exchange the blocks – which is exactly why the algorithm solves for two unknowns R and L rather than one X. The two factors are built one per side, Q_1 from the left graph of L and Z_1 from the right graph of R, and a third transformation re-triangularizes T after they are applied.

When a 2×2 block is involved. The swap is then read off the two deflating subspaces, which is where the **generalized** Sylvester equation enters. In order:

1. Solve $S_{11}R - LS_{22} = \gamma S_{12}$ and $T_{11}R - LT_{22} = \gamma T_{12}$ for R, L; the scaling γ keeps the solution in range.
2. Build Q_1 from a QR factorization of $\begin{pmatrix} -L \\ \gamma I_{n_2} \end{pmatrix}$, and Z_1 from an RQ factorization of $\begin{bmatrix} \gamma I_{n_1}, R \end{bmatrix}$.
3. Apply $Q_1^T (S, T) Z_1$. The coupling is now in the wrong place and T is no longer triangular; restore it two ways – an RQ factorization of the T-part and a QR factorization of the T-part – and keep whichever leaves the smaller residual $(2, 1)$ block of S.
4. Accept the swap only if that residual is below a threshold of order $20\varepsilon_m \|(S_{11}, T_{11}, S_{22}, T_{22})\|$; otherwise the two pairs are too close to separate and the driver reports the pencil as ill-conditioned rather than return a bad ordering.

A block moved k places costs k such swaps, each a fixed-size solve plus $O(n)$ work on the rest of the pencil [6].

6.2. Getting an eigenvector afterwards

The iteration itself produces no eigenvectors: it stops at (S, T, Q, Z). Read the right-eigenvector equation in that basis. Multiplying $Ax = \lambda Bx$ on the left by Q^*, and using Equation 7,

$$S(Z^*x) = \lambda T(Z^*x), \quad \text{so with } y = Z^*x, \quad x = Zy. \tag{23}$$

The Q factor cancels out of a **right** eigenvector, so Z is the only factor needed: solve the triangular pencil (S, T) for y, then rotate back. A **left** eigenvector, $w^*A = \lambda w^*B$, keeps the other factor: putting $u^* = w^*Q^*$ gives $u^*S = \lambda u^*T$, so u comes from the transposed problem and $w = Qu$. That asymmetry is the whole recipe, and it is what LAPACK's `DTGEVC` implements inside `dggev`, which calls it after the Schur form only when the user asks for vectors (`JOBVL` / `JOBVR` = `'V'`; `'N'` never pays for the step) [1], [11].

A 1×1 block. The pair $(\alpha_i, \beta_i) = (S_{ii}, T_{ii})$ is simple, so the diagonal entry $S_{ii} - \lambda T_{ii}$ vanishes and what is left is a triangular solve. Set $y_i = 1$ and $y_j = 0$ for $j > i$ – enough to fix the arbitrary scale, and it satisfies every row below i automatically – then back-substitute rows $i - 1$ down to 1 of $S - \lambda T$. On the example the three directions come out at 8.6×10^{-7} degrees of the reference eigenvectors or closer, with $\|Ax - \lambda Bx\|$ at 5.5×10^{-16} or below. (The reference is itself computed, from the decoupled pencil where the answer is known, so the sub-microdegree part of that angle is the reference's error, not the calculation's.)

The left vector is the mirror image, with one trap. The equation is $u^*(S - \lambda T) = 0$, that is $(S - \lambda T)^* u = 0$, which is **lower** triangular: forward-substitute with $u_i = 1$ and $u_j = 0$ for $j < i$. Conjugating λ before transposing, $\left(S - \overline{\lambda} T\right)^* u = 0$, puts an exact zero on the pivot belonging to the conjugate eigenvalue, and the $0/0$ that follows returns garbage; with the correct form all three vectors land at 1.1×10^{-15} of the library's own left vectors or better.

When a 2×2 block is involved, the block is the unit: solve the 2×2 pencil $(S_2 - \lambda T_2)v = 0$ for the complex pair – never its two diagonal ratios, which are $(1.1 \times 10^{-16}, -1.1 \times 10^{-16})$ on the example – and eliminate the rows above the block as well, since they are coupled to it. The residual comes out at 3.5×10^{-16}.

An infinite eigenvalue needs no new algebra, only the same recipe at $\beta = 0$: the eigen-equation degenerates to $Bx = 0$, so back-substitute on T instead of $S - \lambda T$. At $\delta = 0$ the script returns $\|Bx\| = 1.1 \times 10^{-16}$, $\|Ax\| = 1$, and a direction $0.00°$ from the exact null vector Ue_3.

Measure the residual in the pair. The eigenvector is the same vector whatever test is applied to it, but a ratio-based test destroys its own evidence as β falls – the lesson of section 2 again, now for the friendlier quantity:

δ	$\|Ax - \lambda Bx\|$	$\|\beta Ax - \alpha Bx\|$
10^{-4}	1.30×10^{-12}	1.57×10^{-16}
10^{-8}	5.43×10^{-9}	8.61×10^{-17}
10^{-12}	1.09×10^{-4}	7.89×10^{-17}
10^{-16}	4.90×10^{-1}	4.48×10^{-17}

Table 4: One eigenvector, two residuals, from `scripts/verify_qz.m`. The vector is the one returned for the large eigenvalue $1\frac{\backslash}{\delta}$ of the example; the two columns differ only in how the eigen-equation is written. Read down the second column and the vector appears to fall apart as δ falls; read the third and it sits on the roundoff floor of the pencil instead.

What the library promises. From the `dggev` documentation: a right eigenvector satisfies $Av = \lambda Bv$ and a left one $u^*A = \lambda u^*B$; a complex conjugate pair is stored as two **real** columns, $v_j = VR_{:,j} + iVR_{:,j+1}$ and $v_{j+1} = VR_{:,j} - iVR_{:,j+1}$; and each eigenvector is scaled so its largest component has $|\mathrm{Re}| + |\mathrm{Im}| = 1$, which the script confirms on all three vectors. The scale is a convention and not information, so compare directions and residuals, never magnitudes. In Octave/MATLAB the same call is `[V, D, W] = eig(A, B)`, right vectors in `V` and left ones in `W` ($\|AV - BVD\| = 1.1 \times 10^{-15}$ and $\|W^*A - DW^*B\| = 1.1 \times 10^{-15}$ on the example); SciPy's `scipy.linalg.eig(A, B, left=True)` puts them in one call too.

Two things the vectors cannot do. A repeated eigenvalue has no eigenvector basis at all: on a Jordan block the two vectors returned for the double eigenvalue are parallel ($0.00°$ apart) and the matrix they form has singular values $(1.41, 1, 3.1 \times 10^{-16})$, the last of which is roundoff – one of the two carries no information. And when eigenvalues are merely **clustered**, the honest object is the deflating subspace of the previous section rather than n individual vectors: the leading columns of Q and Z after a reordering describe the cluster without pretending to resolve it, which is what the swaps of this subsection produce.

In code. The pencil of Equation 5 is one call from an eigenvalue list, and the same call works at $\delta = 0$ where $B^{-1}A$ does not exist:

```
[V, D] = eig(A, B);                          % eigenvalues and eigenvectors
[S, T, Q, Z] = qz(A, B);                     % the generalized Schur form
[Ar, Br, Qr, Zr] = ordqz(S, T, Q, Z, 'lhp');  % reorder: stable pairs first
```

(The Octave/MATLAB `ordqz` shown takes the factors from `qz`; MATLAB's form is `ordqz(A, B, keyword)`. In Python, `scipy.linalg.eig(A, B)`, `scipy.linalg.qz(A, B)` and `scipy.linalg.ordqz` expose the same three calls.)

The underlying drivers are LAPACK's `dggev` (eigenvalues, optionally vectors) and `dgges` (Schur form, with ordering). Measured on the example at $\delta = 1/2$, `eig(A, B)` returns $\{2, i, -i\}$ with eigenpair residuals of 7.6×10^{-16}; at $\delta = 0$ it returns $\{i, -i, -\infty\}$. The second input is doing real work in both calls; that is the whole point.

What QZ is not. It is not a way to get eigenvectors for free: they cost a triangular solve and a rotation by Z (right) or Q (left) on top of the iteration, so a driver asked only for eigenvalues never pays for them. It is not a large-sparse method; its cost is dense $O(n^3)$, and a sparse generalized problem needs a Krylov or shift-invert method that never forms S or T. It is not structure-preserving: for a Hamiltonian or symplectic pencil the QZ form is a valid but unphysical decomposition, and the structured methods that keep the pairing of the spectrum are a different subject. The Riccati note here applies QZ where the structure happens to help (`Control/qz-riccati/qz-riccati.pdf`); the symplectic and doubling notes (`Control/symplectic-matrix/symplectic-matrix.pdf`, `Control/sda-riccati/sda-riccati.pdf`) take the structured road. And it is not an accuracy guarantee: section 2 measured the limit that the pencil's own conditioning sets, which no algorithm removes.

7. Sources and further reading

The QZ algorithm, its generalization from QR, and the property this note leans on in section 2 – "particular attention is paid to the degeneracies which result when B is singular. No inversions of B or its submatrices are used" – are Moler and Stewart's [12]. The generalized Schur decomposition, the Hessenberg-triangular reduction, the implicit double-shift step and its bulge chase, and the "QZ step is a QR step on AB^{-1}" reading are Golub and Van Loan, **Matrix Computations**, 4th edition, section 7.7 [4]; the lecture slides of Arbenz state the same decomposition and reduction as the note does [13]. The real double-shift pattern drawn in Figure 5 follows those two sources. The pair output (α_i, β_i), the warning that the quotient may over- or underflow while α and β stay in range, and the $\beta = 0$ convention are LAPACK's `dggev` documentation [1]; the names `qz`, `ordqz`, `dgges` are the MATLAB/Octave and LAPACK interface names for the same three computations. The Givens rotations of the Hessenberg-triangular reduction, and the real driver's double-shift step and its local construction of the shift vector of Equation 16, are LAPACK's `DGGHRD` and `DHGEQZ`, which this note read [2], [3]. The eigenvector step of subsection 6.2 – back-substitution on the triangular pencil and the rotation back by Z or Q – is LAPACK's `DTGEVC`, which `dggev` calls once the Schur form is in hand; its contract (which equations the vectors satisfy, how a complex pair is stored, and the scaling rule) is read from the `dggev` documentation [1], [11]. The reordering of section 6 – adjacent-block swaps and the generalized Sylvester equation – is Kågström's direct method [6]. Balancing before the QZ iteration is Ward [5].

The deflating subspace of section 6 is quoted from Monov and Tsatsomeros, who define it as the subspace pair $AL \subseteq M, BL \subseteq M$ together with the block-triangular reduction it produces [7], and from Oară and Van Dooren, whose basis-matrix form carries an induced **regular** pencil and reduces to the ordinary invariant-subspace definition when one of the two matrices is the identity [8]. The vanishing-subdiagonal certificate $S_{k+1,k} = 0$ is stated in that form by Kågström and Kressner, who attribute it to Stewart and Sun's **Matrix Perturbation Theory**, chapter VI [9], [10] – the textbook itself was not consulted here.

Not consulted, and offered only as pointers for going further: Stewart's **Matrix Algorithms, Volume II** (chapter 4 on the generalized eigenproblem) and the CAREX/DAREX collections of Riccati benchmarks. No generated data was needed; every number in this note is printed by the two Octave scripts beside it.

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